

CONDITIONAL PMFS / PDFS

HOW DOES ONE RANDOM VARIABLE
INFLUENCE ANOTHER?

GIVEN X , SAY SOMETHING ABOUT Y

X, Y BOTH
DISCRETE

$$P(Y=y \mid X=x)$$

CONDITIONAL PMF.

X DISCRETE
 Y CONTINUOUS

$$f_{Y|X}(y \mid X=x)$$

ONE OF N CLASSES
CLASS-
CONDITIONAL
PDF

X, Y BOTH
CONTINUOUS

$$f_{Y|X}(y|x)$$

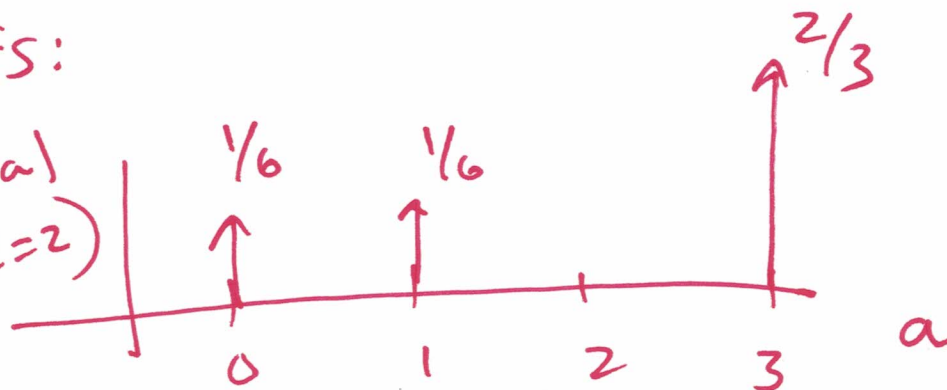
CONDITIONAL PDF.

CONDITIONAL

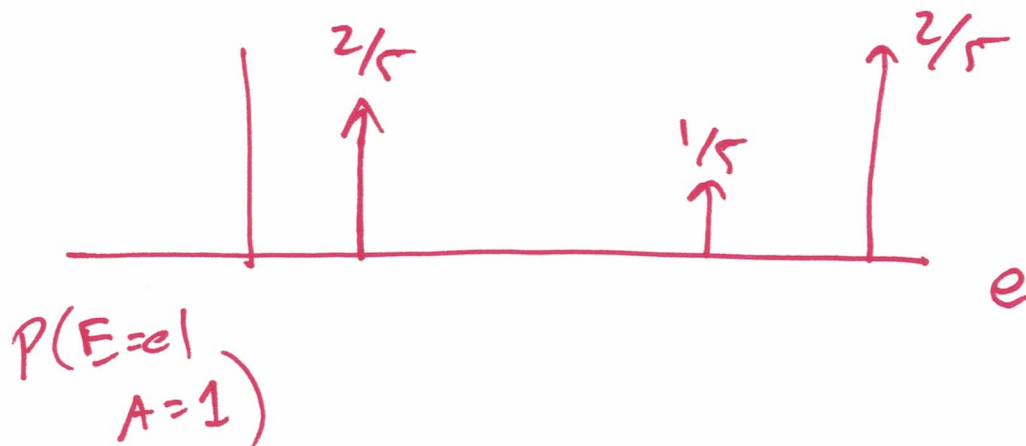
PMFS:

$$P(A | E=2)$$

$$P(A=a | E=2)$$



$$P(E | A=1)$$



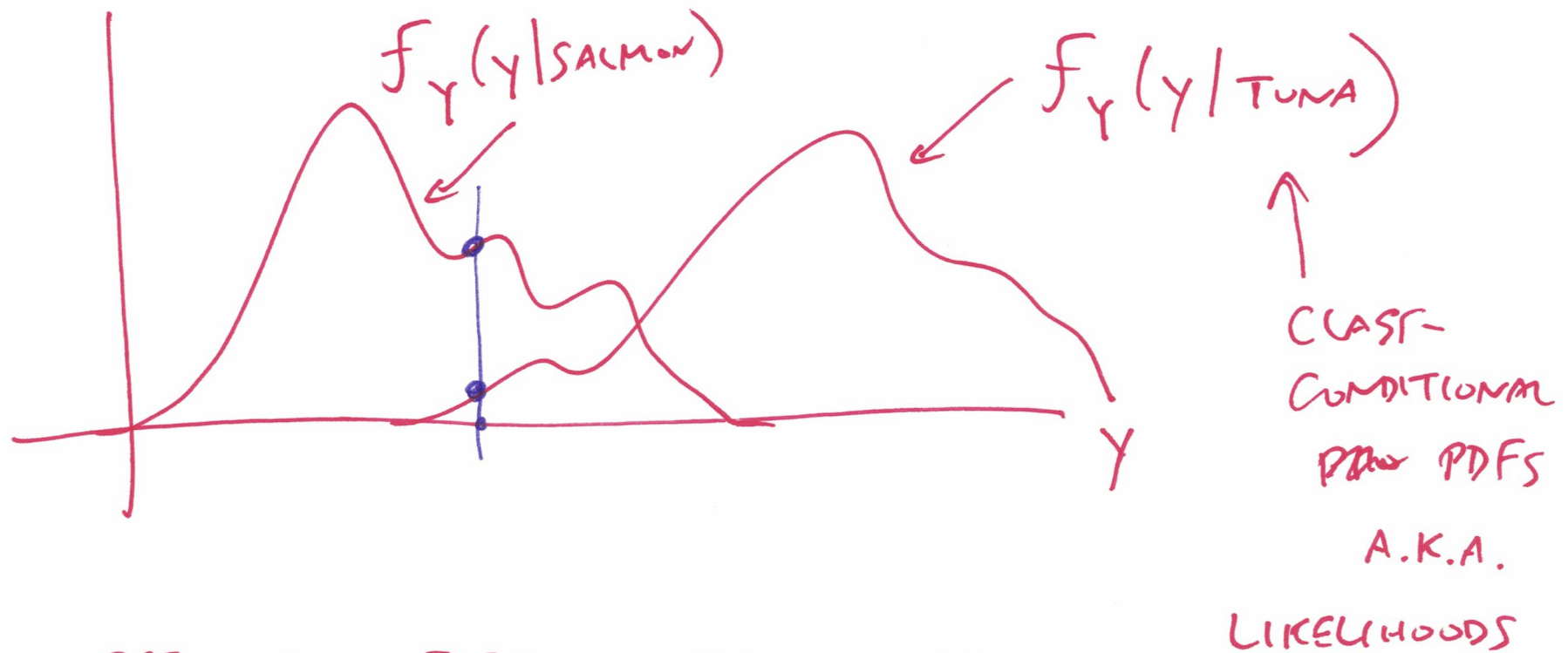
COMMON SITUATION: CLASS-CONDITIONAL
PDF. Y IS A CONTINUOUS RV
THAT DEPENDS ON A DISCRETE VALUE OF X .

EX X IS A LABEL
 Y IS A FEATURE

} ML WAY OF THINKING.

$X \in \{\text{SALMON, TUNA}\}$

$Y = \text{LENGTH OF FISH.}$



NOW SEE A FISH, MEASURE y .

WHICH CLASS IS IT?

SEVERAL WAYS OF DOING IT.

$$\text{IF } f_Y(y|\text{SALMON}) > f_Y(y|\text{TUNA})$$

$\Rightarrow$ CHOOSE SALMON

ELSE $\Rightarrow$ CHOOSE TUNA.

MAXIMUM
LIKELIHOOD
ESTIMATION.

ALTERNATIVE: FORGET ABOUT Y , JUST
 CHOOSE WHICH OF $\frac{P(\text{SALMON})}{P(\text{TUNA})}$ AND
 $\frac{P(\text{TUNA})}{P(\text{SALMON})}$ IS BIGGER.
 ↑ THE PRIOR PROBABILITIES.

RECALL BAYES' RULE:

POSTERIOR PROBABILITY ↓

$$P(\text{SALMON} | Y) = \frac{f_Y(Y | \text{SALMON}) P(\text{SALMON})}{f_Y(Y)}$$

$$= \frac{\sum f_Y(Y | \text{SALMON}) P(\text{SALMON})}{\sum f_Y(Y | \text{TUNA}) P(\text{TUNA}) + \sum f_Y(Y | \text{SALMON}) P(\text{SALMON})}$$

TOTAL
 PROBABILITY
 THEOREM →

MAXIMUM A POSTERIORI (MAP) DECISION RULE:

TAKE WHICHEVER OF

$$P(\text{SALMON} | y)$$

AND $P(\text{TUNA} | y)$ IS BIGGER.

UPDATE PRIORS $\rightarrow$ POSTERIOR BASED ON DATA.

Y SCREAM WHOSE FREQUENCY IS
GAUSSIAN WITH $\mu_Y = 2000 \text{ Hz}$, $\sigma_Y = 100 \text{ Hz}$

R FREQ GAUSSIAN WITH $\mu_R = 1800 \text{ Hz}$, $\sigma_R = 75 \text{ Hz}$.

$$P(R) = 0.85, \quad P(Y) = 0.15$$

WE HEAR A SCREAM AT 1925 Hz .

WHAT SHOULD WE DECIDE ACCORDING TO
BAYESIAN DECISION RULE?

$$P(Y|1925) =$$

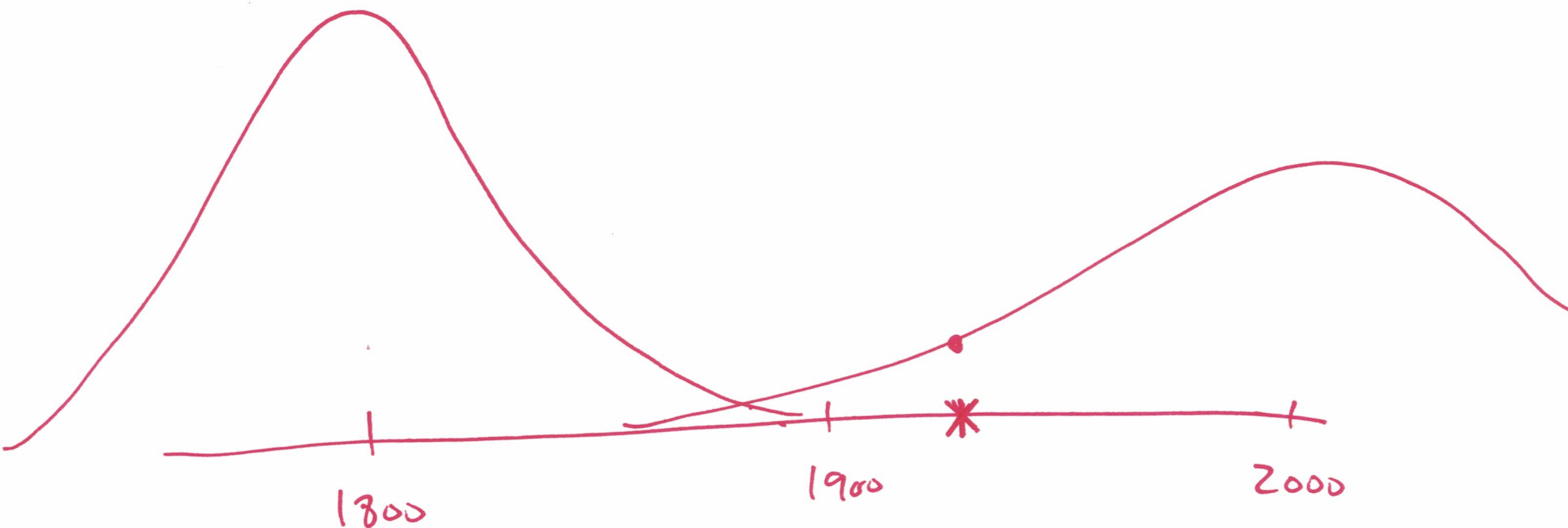
$$\frac{f(1925|Y) P(Y)}{P(1925)}$$

CLASS-COMP. PDF

$$\frac{f(1925|N) P(N)}{P(1925)}$$

WHICH IS
BIGGER?

$$P(N|1925) =$$



$$P(Y) f_Y(1925) = 0.15 \cdot \frac{1}{\sqrt{2\pi} \cdot 100} e^{-\frac{1}{2} \left(\frac{(1925 - \overset{2000}{\cancel{1925}})^2}{(100)^2} \right)} = 0.00045$$

$$P(N) f_N(1925) = 0.85 \cdot \frac{1}{\sqrt{2\pi} \cdot 75} e^{-\frac{1}{2} \left(\frac{(1925 - 1800)^2}{(75)^2} \right)} = \boxed{0.0011}$$

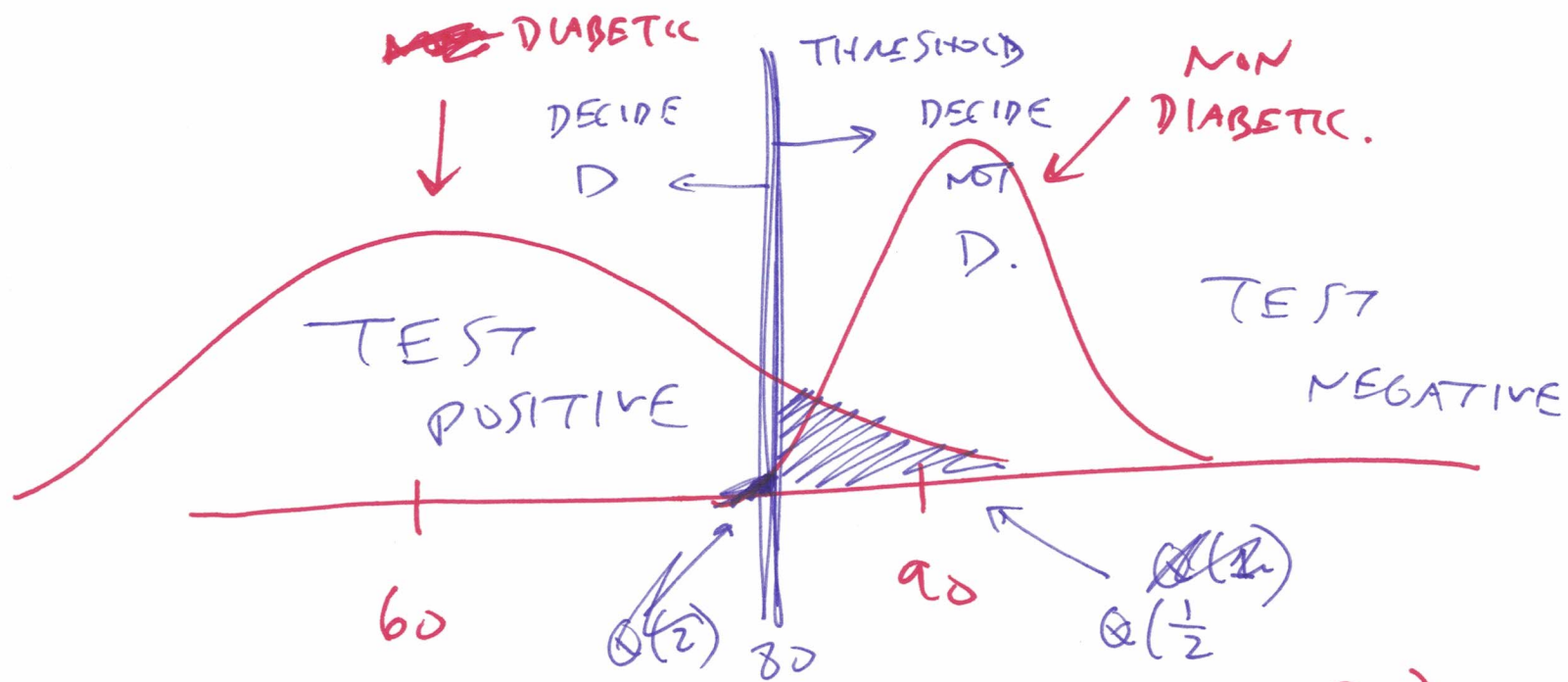
BIGGER

$\Rightarrow$ CHOOSE N .

EX DIABETES TEST PROVIDES A
MEASUREMENT X

NON-DIABETIC: X GAUSSIAN $\mu = 90, \sigma = 20$

DIABETIC X GAUSSIAN $\mu = 60, \sigma = 40$



$$P(D) = 0.1$$

$$P(\text{NOT } D) = 0.9.$$

$$P(D \mid \text{TEST POSITIVE})$$

$$= \frac{P(T \text{ pos} \mid D) P(D)}{P(T \text{ pos} \mid D) P(D) + P(T \text{ pos} \mid \text{NOT } D) P(\text{NOT } D)}$$

